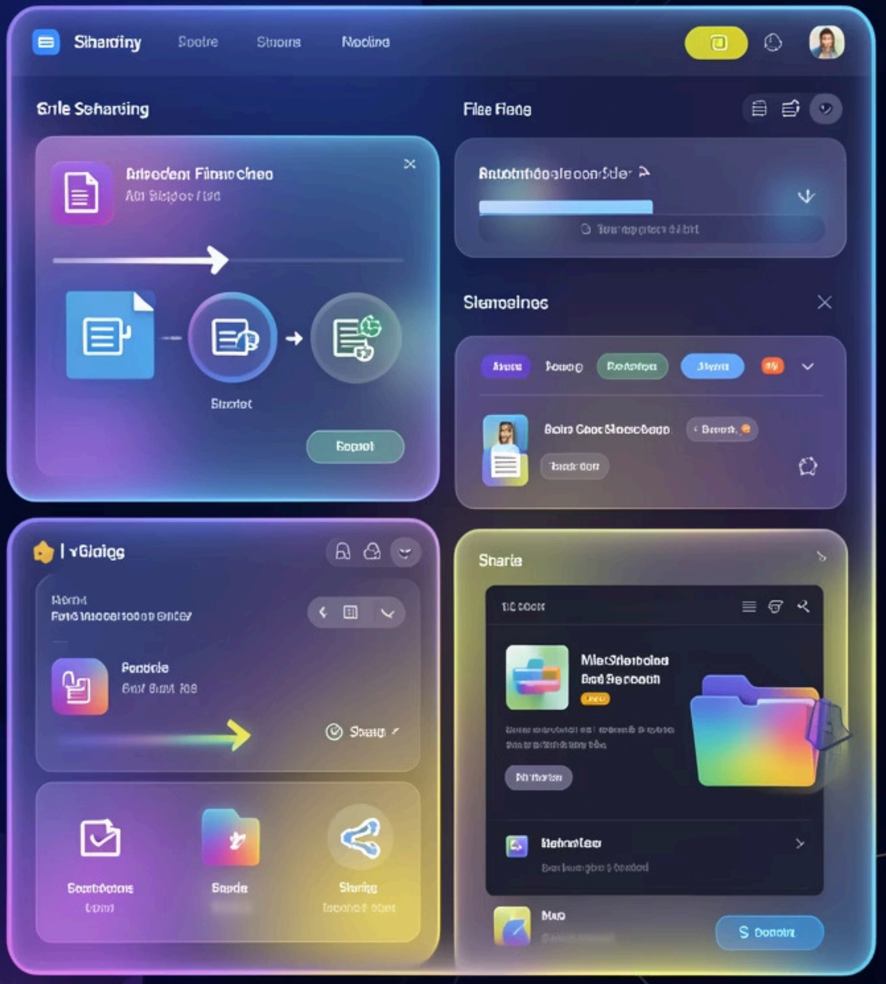


File Sharing Portal using ASP.NET Core MVC

Major Project Presentation

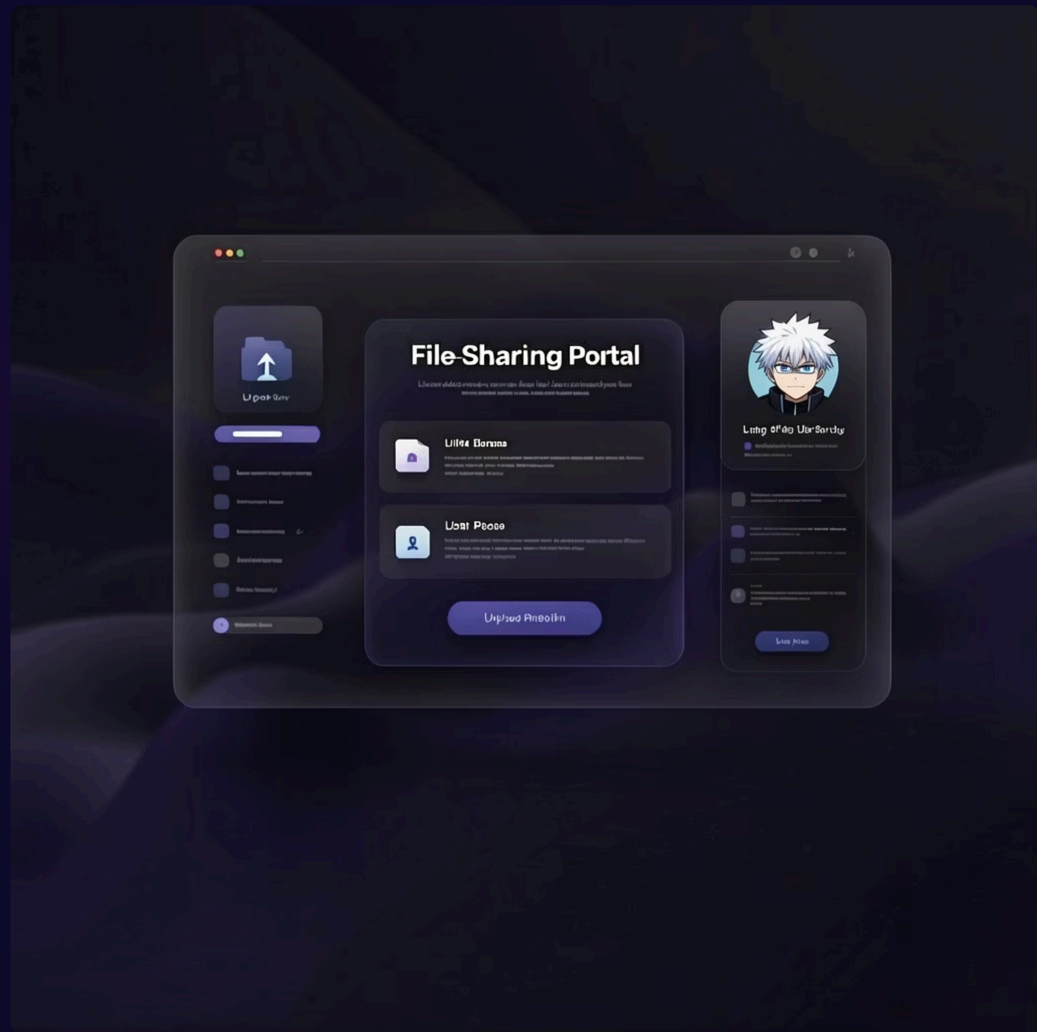
Presented by:

- Sushant Sutar
- Pritam Ghatage
- Shubham Dubal

Main Page: Modern UI Design Prototype

Our main page design focuses on a contemporary user experience, integrating key modern UI elements:

- **Glassmorphism:** Frosted, translucent panels create visual depth and a sense of layered information, making content distinct yet part of a cohesive whole.
- **Subtle 3D Depth:** Carefully applied shadows and slight elevations provide a tactile feel to interactive elements like buttons and cards, enhancing user engagement.
- **Clean & Minimalist Layout:** A streamlined interface reduces clutter, prioritizing essential functions and ensuring ease of navigation for file management.
- **Intuitive Interaction:** These design choices combine to offer an aesthetically pleasing and highly functional platform, making file sharing and management seamless.



Abstract

This project presents a secure, scalable, and modern web-based File Sharing Portal developed using ASP.NET Core MVC. The system allows authenticated users to upload, share, receive, comment, and manage files with strict ownership-based access control.

The platform incorporates modern 3D glassmorphism UI components to enhance user experience while maintaining robust security and scalability principles essential for enterprise applications.

Introduction & Problem Statement

Web-Based System

Secure file management platform accessible through modern browsers

Secure Access

Authentication required for all file operations and data access

Community Sharing

Both private and community-based file distribution capabilities

Existing Platform Limitations

Limited Capacity

Artificial restrictions on file sharing and upload sizes

Weak Ownership Control

Poor permission management and access validation

Outdated UI/UX

Poor user experience and interface design

No Community Module

Lack of structured platform for collaborative sharing

Project Objectives & Core Features

Key Objectives

01

Implement **secure authentication system**

02

Allow **unlimited file uploads**

03

Develop **ownership-based permission control**

04

Provide **community sharing module**

05

Ensure **scalable backend architecture**

Core Features

Secure Login

Authentication required to share and receive files

Unlimited Sharing

No artificial limits on file sharing capacity

Glassmorphism UI

Modern 3D frosted glass interface design

UI Components

Pre-created modern 3D UI components library

Community Section

Dedicated platform for public file sharing

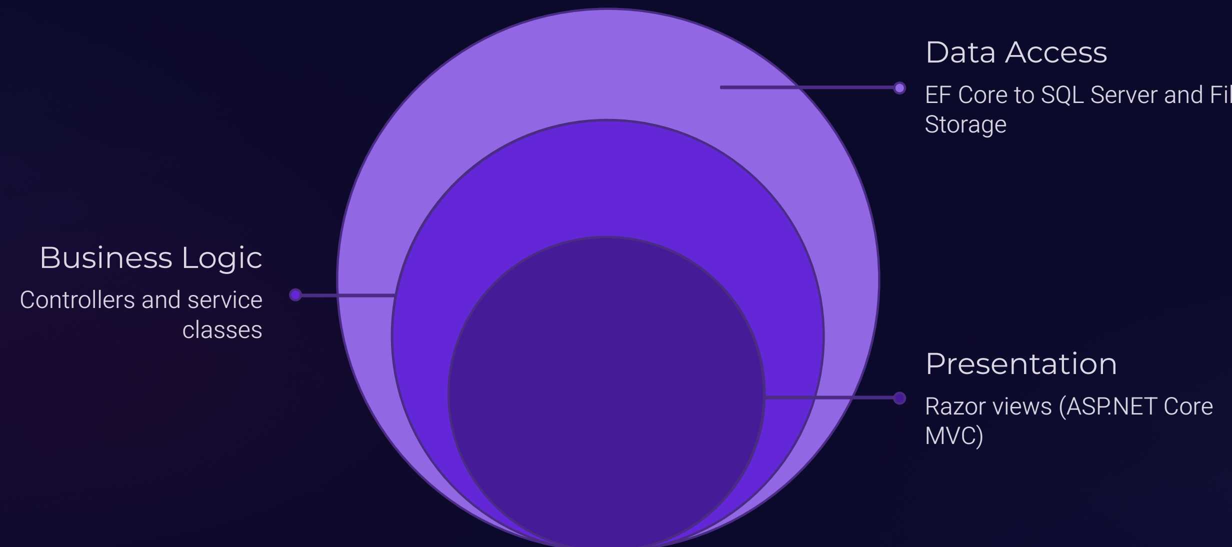
Permission & Access Control System

The system implements granular ownership-based access control where file owners maintain complete control while receivers have limited permissions.

Action	File Owner	File Receiver
Upload	✓ Allowed	▯ Not Allowed
Delete	✓ Allowed	▯ Not Allowed
Modify	✓ Allowed	▯ Not Allowed
Comment	✓ Allowed	✓ Allowed
Download	✓ Allowed	✓ Allowed

Community files follow the same ownership logic — any file shared publicly or in the community section maintains owner privileges for modification and deletion while viewers receive read-only access with commenting capabilities.

System Architecture



Three-Tier Architecture

- **Presentation Layer:** Razor Views with glassmorphism UI
- **Business Logic Layer:** Controllers and service classes
- **Data Access Layer:** Entity Framework Core ORM
- **Storage Layer:** SQL Server Database + File System

Technology Stack

Backend Stack

- ASP.NET Core MVC
- C# Programming
- Entity Framework Core
- SQL Server

Frontend Stack

- Razor Views Engine
- HTML5 & CSS3
- JavaScript
- Glassmorphism UI Components

Security Stack

- Authentication & Authorization
- Password Hashing
- Role-Based Access Control
- Secure File Storage

Database Design & Workflow

Database Schema

Users Table

User profiles, credentials, and authentication data

Files Table

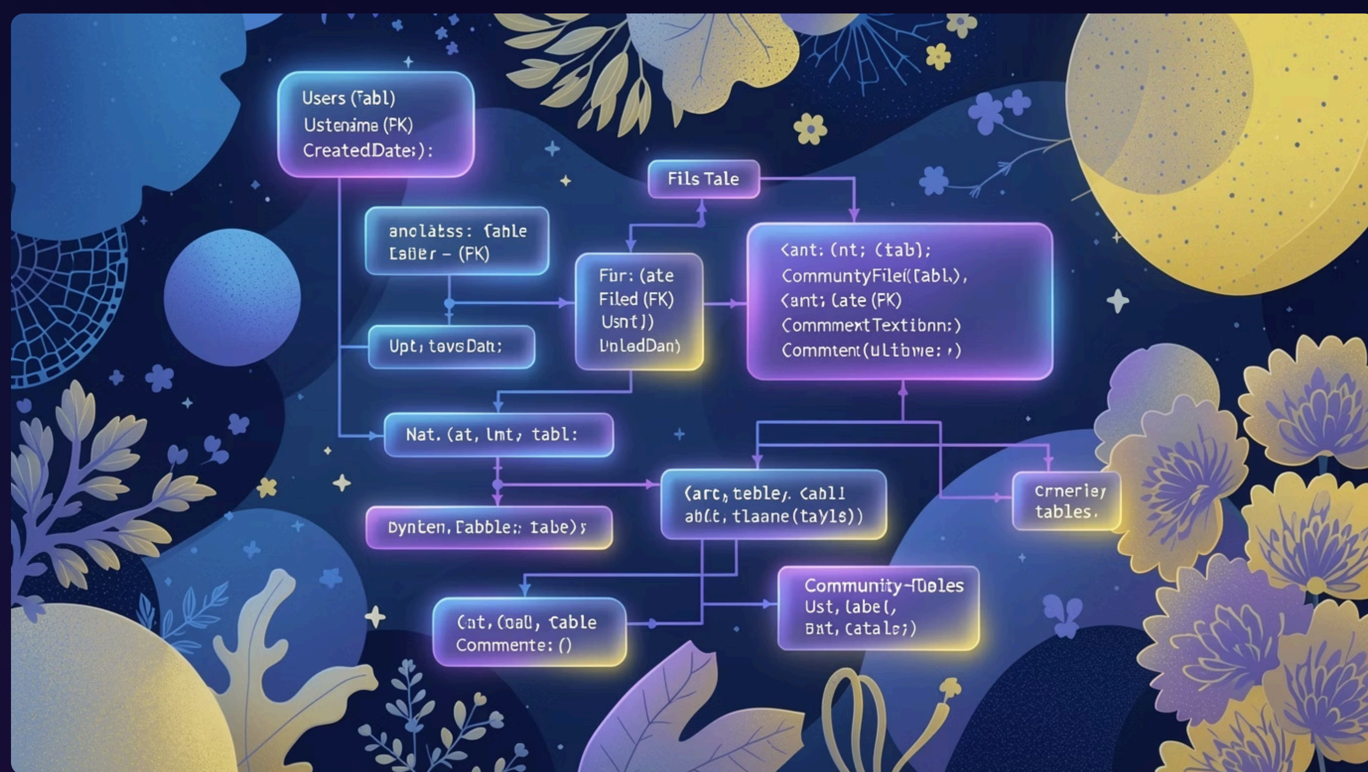
File metadata, storage paths, and ownership tracking

Comments Table

Comment threads linked to specific files

Community Files Table

Public file sharing catalog with access controls



System Workflow

Authentication

User registers or logs in to access the system

File Upload

User uploads file with metadata to server

Share Files

Files shared privately or published to community

Access & Download

Receivers download files and add comments

Permission Validation

System validates ownership before any modification

Security Mechanism & Advantages

Security Mechanism

1

Encrypted passwords using **secure hashing algorithms**

2

Authentication middleware validates **session and user identity**

3

Authorization validation before **every file operation**

4

Secure file storage path management with **encrypted references**

5

Owner-based access verification for **delete and modify operations**

System Advantages

Secure & Reliable

Robust authentication and encryption protect user data

Scalable Architecture

Three-tier design supports growth and increased load

Modern UI Experience

3D glassmorphism design enhances user engagement

Clear Ownership Control

Granular permissions prevent unauthorized access

Academic & Real-World Relevance

Practical application of enterprise development patterns

Future Enhancements & Conclusion

Future Enhancements

Cloud Deployment

Azure integration for elastic scaling and reliability

File Encryption

End-to-end encryption for maximum data security

Real-Time Notifications

WebSocket-based alerts for file activities

Drag & Drop Upload

Intuitive file upload interface

Mobile Responsive

Adaptive design for all devices

Analytics Dashboard

Admin portal with usage statistics

Conclusion

> The File Sharing Portal provides a secure, scalable, and modern solution for file management using ASP.NET Core MVC. With ownership-based permissions, community sharing, and an advanced 3D glassmorphism interface, the system successfully demonstrates both strong backend architecture and modern frontend design principles suitable for real-world enterprise applications.